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Unified performance bounds for generalized selection diversity combining in independent generalized fading channels

Limites unifiées de performance pour la diversité par combinaison avec sélection généralisée dans les canaux indépendants généralisés d'évanouissements

Y. Chen, C. Tellambura, and A. Annamalai*

The cumulative distribution function (cdf) and the moment generating function (mgf) of a linear sum of ordered random variables (RVs) are needed for characterizing the performance of a generalized selection combining (GSC) receiver. Unfortunately, in all but a few cases, the mgf and the cdf either are not in closed form or exist only as multidimensional integrals. Consequently, analyzing the GSC performance in a generalized fading environment is difficult and time-consuming, especially when the number of diversity branches, L, is large. This paper therefore derives new upper and lower bounds for the cdf of the GSC output signal-to-noise ratio (SNR) in generalized fading channels. These bounds are either closed-form or single finite integral expressions. New bounds are also derived for the average symbol error rates (SERs) for a multitude of binary and M-ary digital modulations in a variety of fading channel models that heretofore had resisted simple solutions.

On veut trouver la fonction de distribution cumulative (fdc) et la fonction génératrice des moments (fgm) d'une somme linéaire de variables aléatoires ordonnées. Ces fonctions sont nécessaires pour caractériser la performance d'un récepteur de combinaison par sélection généralisée (CSG). Malheureusement, sauf pour quelques cas simples, le fgm et le fdc n'existent pas en forme exacte ou existent seulement en forme d'intégrales multidimensionnelles. Il est donc long et difficile d'analyser la performance de la CSG dans un environnement d'atténuation généralisée, surtout quand le nombre de branches de diversité, L, est grand. Cette article démontre donc des nouvelles limites supérieures et inférieures pour le fdc du rapport signal-a-bruit de CSG dans un canal d'atténuation généralisée. Ces limites sont sous formes exactes ou sous formes d'intégrale simple. Des nouvelles limites sont également démontrées pour les taux d'erreur moyens de symbole pour une multitude de modulations binaires et M-aires dans une variété de modèles d'évanouissements de canal qui, jusqu'ici, n'avaient pas de solutions simples.

Keywords: diversity, generalized fading channel, generalized selection combining, outage probability

I. Introduction

The generalized selection combining (GSC) diversity scheme has received much attention recently (see [1]–[10] among many others). The ability to capture a significant amount of the transmitted signal energy present in the resolvable multipaths using only a few rake fingers (correlators) is an important consideration for receiver design of wideband code division multiple access (CDMA) and ultra-wideband (UWB) communication systems. Although the maximal-ratio-combining (MRC)-type rake receiver solution (i.e., coherently combining all the L resolvable multipaths) is optimum, it becomes too complex when the number of multipath components, L, is large. This fact motivates the development of suboptimal coherent GSC (GSC(M, L)) techniques [3] in which a subset of $M \leq L$ paths with the highest signal-to-noise ratios (SNRs) is optimally weighted and summed. Clearly, GSC$(1, L)$ and GSC(L, L) are simply the classical selection combining (SC) and MRC receivers. The need for system designers to evaluate the performance of GSC receivers for a large L (say in the order of 20 or more) over a wide variety of fading models is prompted by the emergence of UWB systems and other high-capacity systems [11]–[12].

In this paper, we derive new upper and lower bounds for the outage and the average error rate of GSC receivers. These bounds are *not* based upon the conventional Chernoff bounding approach [13]. Several factors motivate the derivation of these bounds. Despite much recent research on the GSC receiver performance, *simple* analytical results exist only when the diversity branches experience independently and identically distributed (i.i.d.) Rayleigh fading (for example, closed-form expressions for the outage have been published for this case only [14]). It is well known that the moment generating function (mgf) of the GSC output SNR can be used to obtain the average error rates and the outage probabilities. A simple closed-form expression for the mgf exists only for the i.i.d. Rayleigh case[1]. Even for independently and non-identically distributed (i.n.d.) Rayleigh fading statistics, the mgf expression is rather complex [14, Eq. (53)]. Although for i.i.d. Nakagami-m statistics (where m is limited to positive integers) a closed-form mgf expression can be derived [15], it is again rather complex. A novel technique derived in [16] and [17] allows the mgf to be expressed as a single integral for any common fading models such

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[1]That is, the branch SNRs are i.i.d. exponential samples. The general problem of the mgf of a linear sum of order statistics remains an open problem even within the wider mathematical statistics community, where order statistics have been researched since the 1930s.

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as Rice, Weibull and others. Using this technique, Ma and Pasupathy derive the mgf of a GSC receiver in generalized fading channels [9]. However, the outage, which is in fact the cumulative distribution function (cdf), cannot readily be obtained analytically as a Laplace inversion of the mgf, and numerical Laplace inversion essentially leads to a two-dimensional numerical integration. Since the Chernoff bounding approach also requires the mgf, it also fails to be of use for a unified analysis. Moreover, numerically evaluating the GSC receiver performance can be time-consuming for a large L and for non-identical fading environments such as those in UWB communications (as $M!$ terms are needed). Our new outage bounds overcome these difficulties because they can be applied to any fading model and can handle both i.i.d. and i.n.d. statistics.

The paper is organized as follows. Section II develops a set of simple and improved bounds for the outage of a GSC receiver. Section III derives the average error rate bounds from the outage bounds. Section IV provides numerical results, and Section V concludes the paper.

II. Bounds on the outage of GSC in arbitrary fading channels

Consider a GSC system with L branches. The output SNR is given by

$$\gamma_{\text{gsc}} = \sum_{k=L-M+1}^{L} \gamma_{(k)}, \tag{1}$$

where $\gamma_{(1)} \leq \gamma_{(2)} \leq \cdots \leq \gamma_{(L)}$ represents the order statistics obtained by arranging the instantaneous branch SNRs $\{\gamma_k\}$ ($k = 1, \ldots, L$) in an increasing order of magnitude.

Throughout this paper, we will adopt the following notational conventions. For random variable (RV) ξ, the probability density function (pdf), the cdf and the mean are denoted by $f_\xi(x)$, $F_\xi(x)$ and $\overline{\xi}$ respectively. For multichannel reception, $f_{\gamma,i}(x)$ and $F_{\gamma,i}(x)$ denote the pdf and the cdf of the i-th branch SNR with mean value $\overline{\gamma}_i$. We assume that L diversity branches are available for processing and that they are independently distributed (uncorrelated). By a generalized fading channel, we mean two cases: the same $f_\gamma(x)$ characterizes all the branch pdfs but with different average powers (i.e., $\overline{\gamma}_i$ varies with i), or more generally $f_{\gamma,i}(x)$ comes from different distribution families (mixed fading).

A. Stochastic fading channel models

Several statistical models are used to describe the amplitudes of the received signals in different fading scenarios. The Rayleigh distribution is usually used to model the fading signal amplitudes in large cells (macrocellular environments) in the absence of a direct line-of-sight component or tropospheric and ionospheric propagation based on reflection and refraction [18]–[19]. The Ricean distribution can be used in describing the microcellular, picocellular and mobile satellite channels [18]–[19]. Fading signal amplitudes in satellite links that are subject to strong ionospheric scintillation tend to follow the Nakagami-q distribution (also known as the Hoyt distribution) [20]. Recent experimental results have shown that the Weibull distribution is valid for the narrowband path loss at digital enhanced cordless telecommunications (DECT) frequency (1.89 GHz) in an indoor environment [21]. The pdfs, $f_\gamma(x)$, and the cdfs, $F_\gamma(x)$, of the instantaneous branch SNR γ in these fading channels can be found in Table 1.

B. Some basic results from order statistics

A GSC receiver processes order statistics. Here we list some basic results regarding order statistics without proofs. The interested reader is advised to consult [22] for further details. Let $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ be the order statistics obtained from a random i.i.d. sample of size n with pdf $f(x)$ and cdf $F(x)$. Then the pdf of $X_{(r)}$ is given by [22]

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!}\left[F(x)\right]^{r-1}\left[1-F(x)\right]^{n-r} f(x) \tag{2}$$

Table 1
The pdfs, $f_\gamma(x)$, and cdfs, $F_\gamma(x)$, of the instantaneous SNR for several fading channel models

Channel	The pdf and cdf of the instantaneous SNR (γ)
Rayleigh	$f_\gamma(x) = \frac{1}{\overline{\gamma}} e^{-\frac{x}{\overline{\gamma}}}$, $F_\gamma(x) = 1 - e^{-\frac{x}{\overline{\gamma}}}$
Ricean $K \geq 0$	$f_\gamma(x) = \frac{1+K}{\overline{\gamma}} e^{-K-\frac{(1+K)x}{\overline{\gamma}}} I_0\left[2\sqrt{\frac{K(K+1)x}{\overline{\gamma}}}\right]$, $F_\gamma(x) = 1 - Q\left[\sqrt{2K}, \sqrt{\frac{2(K+1)x}{\overline{\gamma}}}\right]$, where $Q\left(\sqrt{2a}, \sqrt{2b}\right) = \int_b^\infty e^{-t-a} I_0(2\sqrt{at})\, dt$
Nakagami-q $-1 \leq b \leq 1$, $b = \frac{1-q^2}{1+q^2}$, $q \geq 0$	$f_\gamma(x) = \frac{1}{\overline{\gamma}\sqrt{1-b^2}} e^{\frac{-x}{(1-b^2)\overline{\gamma}}} I_0\left[\frac{bx}{(1-b^2)\overline{\gamma}}\right]$, $F_\gamma(x) = \sqrt{1-b^2} I_e\left[b, x(1-b^2)\overline{\gamma}\right]$, where $I_e(\frac{V}{U}, U) = \frac{U}{W}\left[Q\left(\sqrt{U+W}, \sqrt{U-W}\right) - Q\left(\sqrt{U-W}, \sqrt{U+W}\right)\right]$, $W = \sqrt{U^2 - V^2}$
Nakagami-m $m \geq 0.5$	$f_\gamma(x) = \frac{1}{\Gamma(m)}\left(\frac{m}{\overline{\gamma}}\right)^m x^{m-1} e^{\frac{-mx}{\overline{\gamma}}}$, $F_\gamma(x) = 1 - \frac{1}{\Gamma(m)}\Gamma\left(m, \frac{mx}{\overline{\gamma}}\right)$, where $\Gamma(a, x) = \int_x^\infty e^{-t} t^{a-1}\, dt$
Weibull	$f_\gamma(x) = \frac{m x^{m-1}}{\beta^m} e^{-\left(\frac{x}{\beta}\right)^m}$, $F_\gamma(x) = 1 - e^{-\left(\frac{x}{\beta}\right)^m}$

for $1 \leq r \leq n$. The joint pdf of $X_{(r)}$ and $X_{(s)}$ ($1 \leq r < s \leq n$) is given by [22]

$$\begin{aligned} f_{X_{(r)},X_{(s)}}(x,y) = &\frac{n!}{(r-1)!(s-r-1)!(n-s)!} \\ &\times [F(x)]^{r-1}[F(y)-F(x)]^{s-r-1} \\ &\times [1-F(y)]^{n-s} f(x) f(y) \end{aligned} \tag{3}$$

for $-\infty \leq x < y < \infty$. The key difficulty that should be kept in mind is that even though $X_1, X_2, \ldots, X_n$ are independent, $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ are not. This fact prevents the development of simple expressions for the mgf and the cdf of a linear sum of order statistics (e.g., (1)), and the issue remains as an open problem even in mathematical statistics literature [22].

C. Bounds for the outage of GSC

Outage is the probability that the output SNR γ_{gsc} falls below a given threshold γ_{th} that depends on the modulation types and other system design requirements [23]. The outage may be defined as

$$P_{\text{out}} = \Pr(\gamma_{\text{gsc}} \leq \gamma_{\text{th}}) = F_{\gamma_{\text{gsc}}}(\gamma_{\text{th}}). \tag{4}$$

This outage is fairly difficult to compute, as γ_{gsc} is a sum of correlated RVs and analytical expressions exist only for several special cases. Consequently, we would like to bound the outage without using the mgf of the GSC output SNR. For this reason, we adapt an idea due to Slimane [24], who develops bounds for a sum of independent lognormal RVs by using the cdf of the maximum.

In order to derive outage bounds that do not require knowledge of the mgf of the combined output SNR, we note that

$$\gamma_{(L)} \leq \gamma_{\text{gsc}} \leq M\gamma_{(L)}. \tag{5}$$

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The above leads to

$$\Pr(M\gamma_{(L)} \le \gamma_{\text{th}}) \le \Pr(\gamma_{\text{gsc}} \le \gamma_{\text{th}}) \le \Pr(\gamma_{(L)} \le \gamma_{\text{th}}). \quad (6)$$

To evaluate the above upper and lower bounds, we need the distribution of $\gamma_{(L)}$, which is well known. Hence, the simple bounds (6) can be readily computed as

$$\prod_{i=1}^{L} F_{\gamma,i}\left(\frac{\gamma_{\text{th}}}{M}\right) \le P_{\text{out}} \le \prod_{i=1}^{L} F_{\gamma,i}(\gamma_{\text{th}}). \quad (7)$$

Equation (7) is exact if $M = 1$, and the bounds should be tight when $M \ll L$ (since the lower bound tends to be quite close to the "exact" curve). Note that these bounds can be applied to any fading model such as Ricean, Nakagami, Weibull and others. However, when M is not small in comparison to L, the above bounds may not be accurate enough for some purposes.

We note that the GSC output SNR can also be bounded using both the largest and the second-largest branch SNRs as

$$\gamma_{(L)} + \gamma_{(L-1)} \le \gamma_{\text{gsc}} \le \gamma_{(L)} + (M-1)\gamma_{(L-1)} \quad \text{for} \quad M \ge 2. \quad (8)$$

Hence, a set of improved bounds for the outage can be obtained as

$$I(M-1, \gamma_{\text{th}}) \le P_{\text{out}} \le I(1, \gamma_{\text{th}}), \quad (9)$$

where both the lower and the upper bounds are special cases of $I(\alpha, \epsilon)$, which is given by

$$\begin{aligned} I(\alpha,\epsilon) &= \Pr(\gamma_{(L)} + \alpha\gamma_{(L-1)} \le \epsilon) \\ &= \int_0^{\frac{\epsilon}{1+\alpha}} \int_x^{\epsilon-\alpha x} f_{\gamma_{(L-1)},\gamma_{(L)}}(x,y)\, dy\, dx. \end{aligned} \quad (10)$$

Using [22, Eq.(2.8.1)], we can show that $I(\alpha, \epsilon)$ can be evaluated as a sum of single finite integrals as

$$\begin{aligned} I(\alpha,\epsilon) = &\sum_{r=1}^{L}\sum_{\substack{s=1\\ s\neq r}}^{L} \int_0^{\frac{\epsilon}{1+\alpha}} \prod_{t=1, t\neq r,s}^{L} F_{\gamma,t}(x) \\ &\times \{F_{\gamma,r}(\epsilon-\alpha x) - F_{\gamma,r}(x)\} f_{\gamma,s}(x)\, dx. \end{aligned} \quad (11)$$

For some fading models, $I(\alpha, \epsilon)$ can even be derived in closed form. However, (11) can also be computed readily for any generalized fading model via numerical integration. These tighter bounds can be useful for many performance studies.

Next we illustrate some special cases of (11).

1. *I.n.d. Rayleigh fading*

Using the pdf and the cdf of the branch SNR in Rayleigh fading (see Table 1), we obtain (11) in closed form as

$$\begin{aligned} I(\alpha,\epsilon) = &\sum_{r=1}^{L}\sum_{\substack{s=1\\ s\neq r}}^{L} \sum_{\substack{b_1+\cdots+b_L=2\\ b_r=b_s=1}}^{L} (-1)^{b_1+\cdots+b_L} \\ &\times \left\{ \frac{1-e^{-\frac{\epsilon\kappa}{1+\alpha}}}{\overline{\gamma}_s\kappa} - e^{-\frac{\epsilon}{\overline{\gamma}_r}} C(b_1,\ldots,b_L) \right\}, \end{aligned} \quad (12)$$

where $b_i = 0$ or 1 for $i = 1, 2, \ldots, L$,

$$\kappa = \frac{b_1}{\overline{\gamma}_1} + \cdots + \frac{b_L}{\overline{\gamma}_L} \quad (13)$$

and

$$C(b_1,\ldots,b_L) = \begin{cases} \dfrac{\epsilon}{(1+\alpha)\overline{\gamma}_s}, & \alpha = \overline{\gamma}_r\kappa - 1, \\[2ex] \dfrac{1-e^{-\frac{\epsilon}{1+\alpha}[\kappa - \frac{1+\alpha}{\overline{\gamma}_r}]}}{\overline{\gamma}_s\left[\kappa - \dfrac{1+\alpha}{\overline{\gamma}_r}\right]}, & \text{otherwise.} \end{cases}$$

2. *I.i.d. fading*

For i.i.d. fading channels, $F_{\gamma,1}(x) = \cdots = F_{\gamma,L}(x) = F_\gamma(x)$ and (11) reduces to

$$\begin{aligned} I(\alpha,\epsilon) = L(L-1) &\int_0^{\epsilon/(1+\alpha)} [F_\gamma(x)]^{L-2} \\ &\times \{F_\gamma(\epsilon-\alpha x) - F_\gamma(x)\} f_\gamma(x)\, dx. \end{aligned} \quad (14)$$

Note that (14) holds for any i.i.d. fading model.

An alternative, interesting proof of (14) is also possible. This proof uses a certain Markov property of order statistics. It is well known that $\gamma_{(L)}$ conditioned on $\gamma_{(L-1)} = x$ is an RV with cdf given by [22]

$$F(y) = \frac{F_\gamma(y) - F_\gamma(x)}{1 - F_\gamma(x)} \quad \text{for} \quad y \ge x. \quad (15)$$

As a result, the quantity $I(\alpha, \epsilon)$ can be expressed as the expectation of a conditional cdf, namely,

$$I(\alpha,\epsilon) = E\left\{ \frac{F_\gamma(\epsilon-\alpha x) - F_\gamma(x)}{1 - F_\gamma(x)} \right\}, \quad (16)$$

where $E\{\cdot\}$ denotes the average over the pdf of x. Using (2) with $n = r + 1 = L$, we find that x is distributed with the pdf

$$f_{\gamma_{(L-1)}}(x) = L(L-1)[F_\gamma(x)]^{L-2}[1 - F_\gamma(x)] f_\gamma(x). \quad (17)$$

Therefore, we readily obtain (14).

3. *I.i.d. Rayleigh fading*

Using Table 1 with (14), we can readily show that $I(\alpha, \epsilon)$ can be evaluated in closed form for i.i.d. Rayleigh fading channels as

$$\begin{aligned} I(\alpha,\epsilon) = L(L-1) &\sum_{r=0}^{L-2} (-1)^r \binom{L-2}{r} \\ &\times \left[\left(\frac{1 - e^{-\frac{(r+2)\epsilon}{(1+\alpha)\overline{\gamma}}}}{r+2} \right) - e^{-\frac{\epsilon}{\overline{\gamma}}} C(r) \right], \end{aligned} \quad (18)$$

where

$$C(r) = \begin{cases} \dfrac{\epsilon}{(1+\alpha)\overline{\gamma}}, & \alpha = 1 + r, \\[2ex] \dfrac{1 - e^{-\frac{(r+1-\alpha)\epsilon}{(1+\alpha)\overline{\gamma}}}}{r+1-\alpha}, & \text{otherwise.} \end{cases}$$

Note that (18) can also be derived directly from (12).

4. *I.i.d. Nakagami-m fading*

If the fading index m is a positive integer value, the cdf of the instantaneous branch SNR in Nakagami-m fading (see Table 1) can be expressed as a finite series

$$F_\gamma(x) = 1 - \exp\left(-\frac{mx}{\overline{\gamma}}\right) \sum_{k=0}^{m-1} \frac{\left[\dfrac{mx}{\overline{\gamma}}\right]^k}{k!}. \quad (19)$$

Noting that the pdf $f_\gamma(x)$ is the derivative of the cdf $F_\gamma(x)$ and integrating (14) by parts, we obtain an alternative expression for $I(\alpha, \epsilon)$ as

$$\begin{aligned} I(\alpha,\epsilon) = &\left[F_\gamma\left(\frac{\epsilon}{1+\alpha}\right) \right]^L + \alpha L \\ &\times \int_0^{\frac{\epsilon}{1+\alpha}} [F_\gamma(x)]^{L-1} f_\gamma(\epsilon - \alpha x)\, dx. \end{aligned} \quad (20)$$

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Table 2
CEP derivative for several common modulations

Modulation format	Negative derivative ($-P_e(\gamma)$)
BPSK ($a=1$) C-BFSK ($a=\frac{1}{2}$)	$\frac{\sqrt{a}}{2}\frac{e^{-a\gamma}a}{\sqrt{\pi\gamma}}$
NC-BFSK ($a=\frac{1}{2}$) BDPSK ($a=1$)	$\frac{a}{2}e^{-a\gamma}$
QPSK ($q=a=\frac{1}{2}$) MSK ($q=a=\frac{1}{2}$) M_c-QAM ($a=\frac{3}{2(M_c-1)}$, $q=1-\frac{1}{\sqrt{M_c}}$)	$2q\sqrt{a}\frac{e^{-a\gamma}}{\sqrt{\pi\gamma}}-2q^2\sqrt{a}\frac{erfc(\sqrt{a\gamma})e^{-a\gamma}}{\sqrt{\pi\gamma}}$
CDE-BPSK	$\frac{e^{-\gamma}}{\sqrt{\pi\gamma}}-1.0\frac{erfc(\sqrt{\gamma})e^{-\gamma}}{\sqrt{\pi\gamma}}$
MDPSK	$\frac{1}{\pi}\int_0^{\pi-\frac{\pi}{M}}g(\theta)e^{-\gamma g(\theta)}d\theta$, where $g(\theta)=\frac{\sin^2(\frac{\pi}{M})}{1+\cos(\frac{\pi}{M})\cos\theta}$
MPSK	$\frac{1}{\pi}\int_0^{\pi-\frac{\pi}{M}}h(\theta)e^{-\gamma h(\theta)}d\theta$, where $h(\theta)=\frac{\sin^2(\frac{\pi}{M})}{\sin^2\theta}$

Table 3
$U(\alpha)$ in (23) of several digital modulations in i.i.d. Rayleigh fading channels

Modulation format	$U(\alpha)$ in (23)
BPSK, C-BFSK	$\frac{\sqrt{a}}{2}\sum_{r=0}^{L}(-1)^r\binom{L}{r}\left(a+\frac{\alpha r}{\overline{\gamma}}\right)^{-\frac{1}{2}}$
NC-BFSK, BDPSK	$\frac{a}{2}\sum_{r=0}^{L}(-1)^r\binom{L}{r}\left(a+\frac{\alpha r}{\overline{\gamma}}\right)^{-1}$
QPSK, MSK M_c-QAM	$\sum_{r=0}^{L}(-1)^r\binom{L}{r}\left[2q\sqrt{\frac{a}{a+\frac{\alpha r}{\overline{\gamma}}}}-\frac{4q^2}{\pi}K_1\left(\frac{\alpha r}{a\overline{\gamma}}\right)\right]$, where $K_1(b)$ is defined as (35)
CDE-BPSK	$\sum_{r=0}^{L}(-1)^r\binom{L}{r}\left[\left(1+\frac{\alpha r}{\overline{\gamma}}\right)^{-\frac{1}{2}}-\frac{2}{\pi}K_1\left(\frac{\alpha r}{\overline{\gamma}}\right)\right]$, where $K_1(b)$ is defined as (35)
M_c-DPSK	$\frac{1}{\pi}\sum_{r=0}^{L}(-1)^r\binom{L}{r}K_2\left(\frac{\alpha r}{\overline{\gamma}}\right)$, where $K_2(b)$ is defined as (38)
M_c-PSK	$\frac{1}{\pi}\sum_{r=0}^{L}(-1)^r\binom{L}{r}K_3\left(\frac{\alpha r}{\overline{\gamma}}\right)$, where $K_3(b)$ is defined as (40)

Using (19), we obtain $I(\alpha,\epsilon)$ for the i.i.d. Nakagami-m fading as

$$\begin{aligned} I(\alpha,\epsilon) = &\left[1-e^{-\frac{\epsilon m}{(1+\alpha)\overline{\gamma}}}\sum_{k=0}^{m-1}\frac{1}{k!}\left(\frac{\epsilon m}{(1+\alpha)\overline{\gamma}}\right)^k\right]^L \\ &+\frac{\alpha L e^{\frac{\epsilon m}{\overline{\gamma}}}}{(m-1)!}\sum_{i=0}^{L-1}\binom{L-1}{i}\sum_{j=0}^{m-1}\binom{m-1}{j}(-1)^{i+j} \\ &\times\alpha^j\left(\frac{\epsilon m}{\overline{\gamma}}\right)^{m-j-1}\sum_{k=0}^{(m-1)i}\beta_k(i)C(k,i), \end{aligned} \tag{21}$$

where β_n can be computed recursively as

$$\beta_n(i)=\begin{cases}1, & n=0,\\ \frac{1}{n}\sum_{k=1}^{n}\frac{k(i+1)-n}{k!}\beta_{n-k}(i) & n\neq 0,\end{cases}$$

and

$$C(k,i)=\begin{cases}\frac{1}{k+1}\left(\frac{\epsilon m}{(1+\alpha)\overline{\gamma}}\right)^k, & \alpha=i,\\ \frac{1}{(r-\alpha)^{k+l}}\gamma\left(k+l,\frac{\epsilon m(i-\alpha)}{(1+\alpha)\overline{\gamma}}\right), & \text{otherwise},\end{cases}$$

where $\gamma(a,x)$ is the incomplete gamma function defined as $\gamma(a,x)=\int_0^x t^{a-1}e^t\,dt$, which is available in standard numerical softwares, such as Maple V and MATLAB. As expected, when $m=1$, (21) reduces to (18) for the i.i.d. Rayleigh fading channel.

III. Error rate bounds from the outage bounds

Conventionally, the average error rate is obtained by integrating the conditional error probability, $P_e(\gamma)$, over the pdf of γ_{gsc}. We, however, express the average error probability in terms of the cdf of γ_{gsc}. This can readily be done using the integration-by-parts method:

$$\begin{aligned}\overline{P}_e &= \int_0^\infty P_e(\gamma)f_{\gamma_{\text{gsc}}}(\gamma)\,d\gamma \\ &= \int_0^\infty -P_e'(\gamma)F_{\gamma_{\text{gsc}}}(\gamma)\,d\gamma,\end{aligned} \tag{22}$$

where $P_e'(\gamma)=\dfrac{d}{d\gamma}P_e(\gamma)$ denotes the derivative of the conditional error rate. Such derivatives are given in Table 2. We can use (22) with the outage bounds derived in Section II to derive the simple and the improved error rate bounds of various digital modulations in generalized fading channels.

Combining (7) with (22), we can readily obtain the simple error rate bounds, which are given by

$$U\left(\frac{1}{M}\right)\le\overline{P}_e\le U(1), \tag{23}$$

where

$$U(\alpha)=\int_0^\infty -P_e'(\gamma)\prod_{i=1}^{L}F_{\gamma,i}(\alpha\gamma)\,d\gamma. \tag{24}$$

This can now be applied to any modulation format and any fading model. For Rayleigh fading channels, (24) can be obtained as

$$U(\alpha)=\sum_{b_1+\cdots+b_L=0}^{L}(-1)^{b_1+\cdots+b_L}\int_0^\infty -P_e'(\gamma)e^{-\gamma\kappa}\,d\gamma, \tag{25}$$

where $b_i=0$ or 1 for $i=1,\ldots,L$ and κ is defined as (13). Using the expressions for $-P_e'(\gamma)$ in Table 2, we can readily derive the closed-form expressions for $U(\alpha)$ of different digital modulations in Rayleigh fading channels. Such results can be found in Table 3 for i.i.d. Rayleigh fading channels. For other fading channels, numerical integration can be readily used to obtain the simple bounds.

The above error rate bounds can indeed be improved by combining (9) and (22). Thus an improved set of bounds is derived as

$$V(M-1)\le\overline{P}_e\le V(1), \tag{26}$$

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where

$$V(\alpha) = \int_0^\infty -P_e'(\gamma) I(\alpha,\gamma)\, d\gamma$$
$$= \sum_{r=1}^{L} \sum_{\substack{s=1\\ s\neq r}}^{L} \int_0^\infty -P_e'(\gamma) \int_0^{\frac{\gamma}{1+\alpha}} \prod_{t=1, t\neq r,s}^{L} F_{\gamma,t}(x)$$
$$\times \{F_{\gamma,r}(\gamma-\alpha x) - F_{\gamma,r}(x)\} f_{\gamma,s}(x)\, dx\, d\gamma. \quad (27)$$

Note that (26) holds for any fading model. Next, we derive the closed-form expressions for $V(\alpha)$ of a broad class of binary and M-ary digital modulations in Rayleigh fading. Of course, our approach can be used for any fading channel. However, for brevity, we do not develop such results here.

A. Binary phase-shift keying (BPSK) and coherent binary frequency-shift keying (C-BFSK)

BPSK has not only proven to be a very effective modulation scheme for low-frequency radio, but has also been used for the reverse link in the CDMA2000 standard for its simple implementation and high power efficiency. Substituting the corresponding $P_e'(\gamma)$ in Table 2 and $I(\alpha,\epsilon)$ from (12) into (27), we readily obtain $V(\alpha)$ for BPSK and C-BFSK in i.n.d. Rayleigh fading as

$$V(\alpha) = \frac{\sqrt{a}}{2} \sum_{r=1}^{L} \sum_{\substack{s=1\\ s\neq r}}^{L} \sum_{\substack{b_1+\cdots+b_L=2\\ b_r=b_s=1}}^{L} (-1)^{b_1+\cdots+b_L}$$
$$\times \left\{ \frac{1}{\overline{\gamma}_s \kappa} \left[\frac{1}{\sqrt{a}} - \left(\sqrt{a} + \frac{\kappa}{1+\alpha}\right)^{-1} \right] - D_1(b_1,\ldots,b_L) \right\}, \quad (28)$$

where κ is defined as (13) and

$$D_1(b_1,\ldots,b_L) = \begin{cases} \dfrac{1}{2(1+\alpha)\overline{\gamma}_s \left(a + \frac{1}{\overline{\gamma}_r}\right)^{\frac{3}{2}}}, & \alpha = \overline{\gamma}_r\kappa - 1, \\ \dfrac{\left(a+\frac{1}{\overline{\gamma}_r}\right)^{-\frac{1}{2}} - \left[a + \frac{\kappa}{1+\alpha}\right]^{-\frac{1}{2}}}{\overline{\gamma}_s \left[\kappa - \frac{1+\alpha}{\overline{\gamma}_r}\right]}, & \text{otherwise.} \end{cases}$$

For the i.i.d. Rayleigh fading, $\overline{\gamma}_1 = \cdots = \overline{\gamma}_L = \overline{\gamma}$, and thus (28) reduces to

$$V(\alpha) = \frac{\sqrt{a}}{2} H(\alpha, a), \quad (29)$$

where

$$H(\alpha,g) = \int_0^\infty \frac{e^{-g\gamma}}{\sqrt{\pi\gamma}} I(\alpha,\gamma)\, d\gamma$$
$$= L(L-1) \sum_{r=0}^{L-2} (-1)^r \binom{L-2}{r}$$
$$\times \left[\frac{1}{r+2} \left(\frac{1}{\sqrt{g}} - \frac{1}{\sqrt{g + \frac{r+2}{(1+\alpha)\overline{\gamma}}}} \right) - D_1(r) \right], \quad (30)$$

where

$$D_1(r) = \begin{cases} \dfrac{1}{2(1+\alpha)\overline{\gamma}\left(g + \frac{1}{\overline{\gamma}}\right)^{\frac{3}{2}}}, & \alpha = 1+r, \\ \dfrac{\left(g+\frac{1}{\overline{\gamma}}\right)^{-\frac{1}{2}} - \left(g + \frac{r+2}{(1+\alpha)\overline{\gamma}}\right)^{-\frac{1}{2}}}{r+1-\alpha}, & \text{otherwise.} \end{cases}$$

B. Non-coherent BFSK (NC-BFSK) and binary differential PSK (BDPSK)

Following the same approach as the above, we obtain the improved bounds for the average BER of NC-BFSK and BDPSK in i.n.d. Rayleigh fading as

$$V(\alpha) = \frac{\sqrt{a}}{2} \sum_{r=1}^{L} \sum_{\substack{s=1\\ s\neq r}}^{L} \sum_{\substack{b_1+\cdots+b_L=2\\ b_r=b_s=1}}^{L} (-1)^{b_1+\cdots+b_L}$$
$$\times \left\{ \frac{1}{\overline{\gamma}_s \kappa} \left[\frac{1}{a} - \left(a + \frac{\kappa}{1+\alpha}\right)^{-1} \right] - D_2(b_1,\ldots,b_L) \right\}, \quad (31)$$

where κ is defined as (13) and

$$D_2(b_1,\ldots,b_L) = \begin{cases} \dfrac{1}{(1+\alpha)\overline{\gamma}_s \left(a + \frac{1}{\overline{\gamma}_r}\right)^2}, & \alpha = \overline{\gamma}_r\kappa - 1, \\ \dfrac{\left(a+\frac{1}{\overline{\gamma}_r}\right)^{-1} - \left[a + \frac{\kappa}{1+\alpha}\right]^{-1}}{\overline{\gamma}_s \left[\kappa - \frac{1+\alpha}{\overline{\gamma}_r}\right]}, & \text{otherwise.} \end{cases}$$

Similarly, for i.i.d. Rayleigh fading, where $\overline{\gamma}_1 = \cdots = \overline{\gamma}_L = \overline{\gamma}$, (31) reduces to

$$V(\alpha) = \frac{a}{2} G(\alpha, a), \quad (32)$$

where

$$G(\alpha,g) = \int_0^\infty e^{-g\gamma} I(\alpha,\gamma)\, d\gamma$$
$$= L(L-1) \sum_{r=0}^{L-2} (-1)^r \binom{L-2}{r}$$
$$\times \left[\frac{1}{r+2} \left(\frac{1}{g} - \frac{1}{g + \frac{r+2}{(1+\alpha)\overline{\gamma}}} \right) - D_2(r) \right], \quad (33)$$

where

$$D_2(r) = \begin{cases} \dfrac{1}{(1+\alpha)\overline{\gamma}\left(g + \frac{1}{\overline{\gamma}}\right)^2}, & \alpha = 1+r, \\ \dfrac{\left(g+\frac{1}{\overline{\gamma}}\right)^{-1} - \left(g + \frac{2+r}{1+\alpha}\right)^{-1}}{r+1-\alpha}, & \text{otherwise.} \end{cases}$$

For brevity, we consider only the i.i.d. Rayleigh fading in the following subsections.

C. Quaternary PSK (QPSK), minimum shift keying (MSK) and M_c-quadrature amplitude modulation (QAM)

For QPSK, MSK and M_c-QAM, using the corresponding $P_e'(\gamma)$ in Table 1 and the expression for $I(\alpha,\epsilon)$ from (18) with (27) and changing the order of integration, we obtain a closed-form expression for $V(\alpha)$ in the i.i.d. Rayleigh fading as

$$V(\alpha) = 2q\sqrt{a} H(\alpha,a) - \frac{4q^2\sqrt{a}}{\pi} \int_0^{\frac{\pi}{2}} H\left(\alpha, \frac{a(1+\sin^2\theta)}{\sin^2\theta}\right) d\theta$$
$$= 2q\sqrt{a} H(\alpha,a) - \frac{4q^2}{\pi} L(L-1) \sum_{r=0}^{L-2} (-1)^r \binom{L-2}{r}$$
$$\times \left\{ \frac{1}{r+2} \left[\frac{\pi}{4} - K_1\left(\frac{r+2}{a\overline{\gamma}(1+\alpha)} \right) \right] - D_3(r) \right\}, \quad (34)$$

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where $H(\alpha, g)$ is defined as (30),

$$K_1(b) = \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\left(\frac{1+\sin^2\theta}{\sin^2\theta}\right) + b}} = \frac{\sin^{-1}\sqrt{\frac{1+b}{2+b}}}{\sqrt{1+b}} \tag{35}$$

and

$$D_3(r) = \begin{cases} \dfrac{\sqrt{\frac{a\overline{\gamma}}{1+a\overline{\gamma}}}\sin^{-1}\sqrt{\frac{1+a\overline{\gamma}}{1+2a\overline{\gamma}}} - \frac{a\overline{\gamma}}{1+2a\overline{\gamma}}}{2(1+\alpha)(1+a\overline{\gamma})}, & \alpha = 1+r, \\ \dfrac{K_1\left(\frac{1}{a\overline{\gamma}}\right) - K_1\left(\frac{r+2}{a\overline{\gamma}(1+\alpha)}\right)}{r+1-\alpha}, & \text{otherwise.} \end{cases}$$

D. Coherently detected differentially encoded BPSK (CDE-BPSK)

Similarly for CDE-BPSK, the $V(\alpha)$ in (27) can also be written in terms of $H(\alpha, a)$ from (30) and $K_1(b)$ from (35). After some algebraic manipulation, we obtain $V(\alpha)$ in closed form as

$$\begin{aligned} V(\alpha) &= H(\alpha,1) - \frac{2}{\pi}\int_0^{\frac{\pi}{2}} H\left(\alpha, \frac{1+\sin^2\theta}{\sin^2\theta}\right) d\theta \\ &= H(\alpha,1) - \frac{2}{\pi}L(L-1)\sum_{r=0}^{L-2}(-1)^r\binom{L-2}{r} \\ &\quad \times \left[\frac{1}{r+2}\left(\frac{\pi}{4} - K_1\left(\frac{r+2}{\overline{\gamma}(1+\alpha)}\right) - D_3(r)\right)\right], \end{aligned} \tag{36}$$

where $K_1(b)$ is defined in (35).

E. M_c differential PSK (DPSK)

Several second-generation North American and Japanese digital cellular standards adopt $\pi/4$-DQPSK as their modulation scheme [25]. As above, we may write $V(\alpha)$ in (27) in terms of $G(\alpha, g)$ from (33) for M_c-DPSK. Applying [26, Eq.(2.554)], we obtain an improved bound for the average SER of M_c-DPSK:

$$\begin{aligned} V(\alpha) &= \frac{1}{\pi}\int_0^{\pi-\frac{\pi}{M_c}} \frac{\sin^2\left(\frac{\pi}{M_c}\right)}{1+\cos\left(\frac{\pi}{M_c}\right)\cos\theta} \\ &\quad \times G\left(\alpha, \frac{\sin^2\left(\frac{\pi}{M_c}\right)}{1+\cos\left(\frac{\pi}{M_c}\right)\cos\theta}\right) d\theta \\ &= \frac{L(L-1)}{\pi}\sum_{r=0}^{L-2}(-1)^r\binom{L-2}{r} \\ &\quad \times \left\{\frac{1}{r+2}\left[\frac{\pi(M_c-1)}{M_c} - K_2\left(\frac{r+2}{\overline{\gamma}(1+\alpha)}\right)\right] - D_4(r)\right\}, \end{aligned} \tag{37}$$

where

$$\begin{aligned} K_2(b) &= \int_0^{\pi-\frac{\pi}{M_c}} \frac{g(\theta)}{g(\theta)+b}\, d\theta \\ &= \frac{2\sin\frac{\pi}{M_c}\tan^{-1}\left[\sqrt{\frac{\sin^2\frac{\pi}{M_c}+b\left(1-\cos\frac{\pi}{M_c}\right)}{\sin^2\frac{\pi}{M_c}+b\left(1+\cos\frac{\pi}{M_c}\right)}}\tan\left(\frac{\pi}{2}-\frac{\pi}{2M_c}\right)\right]}{\sqrt{\sin^2\frac{\pi}{M_c}+2b+b^2}} \end{aligned} \tag{38}$$

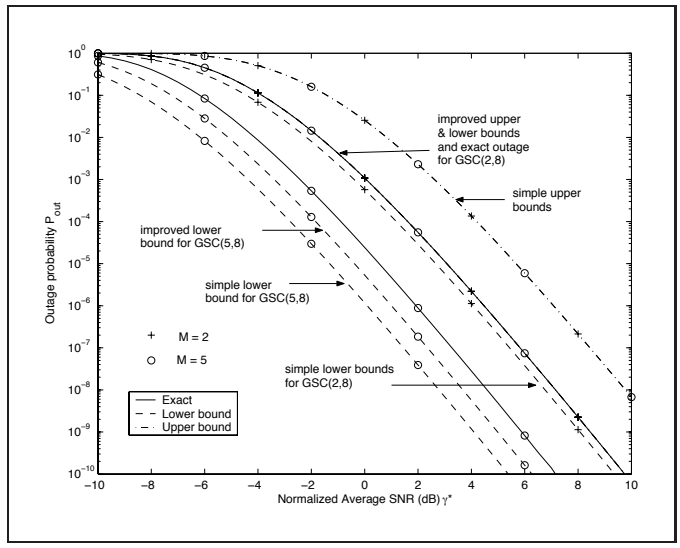

Figure 1: *Outage probability for GSC$(2,8)$ and GSC$(5,8)$ in i.i.d. Rayleigh fading; $\gamma^* = \overline{\gamma}/\gamma_{th}$.*

and

$$D_4(r) = \begin{cases} \dfrac{\overline{\gamma}\left[\frac{\overline{\gamma}\sin\frac{2\pi}{M_c}}{2(1+\overline{\gamma})} + \frac{1+\overline{\gamma}}{\overline{\gamma}}K_2\left(\frac{1}{\overline{\gamma}}\right)\right]}{(1+\alpha)\left(1+2\overline{\gamma}+\overline{\gamma}^2\sin^2\frac{\pi}{M_c}\right)}, & \alpha = 1+r, \\ \dfrac{K_2\left(\frac{1}{\overline{\gamma}}\right) - K_2\left(\frac{r+2}{\overline{\gamma}(1+\alpha)}\right)}{r+1-\alpha}, & \text{otherwise.} \end{cases}$$

As expected, when $M_c = 2$, (37) reduces to (31) for binary DPSK.

F. M_c-PSK

Besides BPSK and QPSK, 8PSK is another commonly used PSK modulation scheme which has been adopted in some 2.5G wireless communication systems such as Enhanced Data for Global Evolution (EDGE) [25]. Substituting the corresponding $P_e'(\gamma)$ and (18) into (27) and applying [26, Eq. (2.551)], we obtain a closed-form expression for $V(\alpha)$ in (27):

$$\begin{aligned} V(\alpha) &= \frac{1}{\pi}\int_0^{\pi-\frac{\pi}{M_c}} \frac{\sin^2\left(\frac{\pi}{M_c}\right)}{\sin^2\theta}\, G\left(\alpha, \frac{\sin^2\left(\frac{\pi}{M_c}\right)}{\sin^2\theta}\right) d\theta \\ &= \frac{L(L-1)}{\pi}\sum_{r=0}^{L-2}(-1)^r\binom{L-2}{r} \\ &\quad \times \left\{\frac{1}{r+2}\left[\frac{\pi(M_c-1)}{M_c} - K_3\left(\frac{r+2}{\overline{\gamma}(1+\alpha)}\right)\right] - D_5(r)\right\}, \end{aligned} \tag{39}$$

where

$$\begin{aligned} K_3(b) &= \int_0^{\pi-\frac{\pi}{M_c}} \frac{h(\theta)}{h(\theta)+b_2}\, d\theta \\ &= \frac{\sin\frac{\pi}{M_c}\tan^{-1}\left[\frac{\sqrt{\sin^2\frac{\pi}{M_c}+b}}{\sin\frac{\pi}{M_c}}\tan\left(\pi-\frac{\pi}{M_c}\right)\right]}{\sqrt{\sin^2\frac{\pi}{M_c}+b}} \end{aligned} \tag{40}$$

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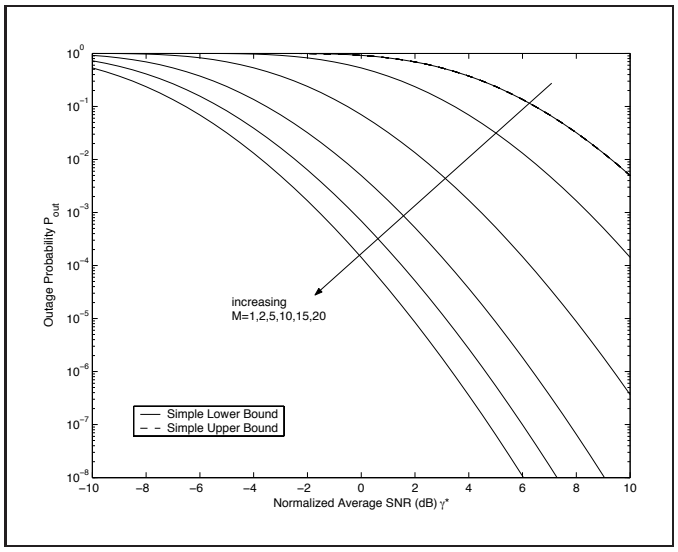

Figure 2: *Simple outage probability bounds for GSC$(M, 30)$ in mixed fading; $\gamma^* = \bar{\gamma}/\gamma_{th}$.*

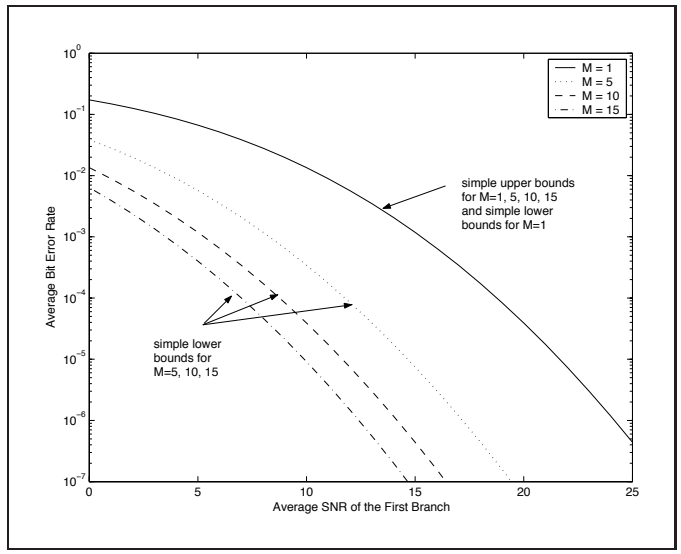

Figure 3: *Simple bounds for the average bit error rate of BPSK with GSC$(M, 15)$ receiver in i.n.d. Nakagami-m fading; $m = 0.75$.*

and

$$D_5(r) = \begin{cases} \dfrac{K_3\left(\dfrac{1}{\overline{\gamma}}\right) - \dfrac{\overline{\gamma}\sin\dfrac{\pi}{M_c}\cos\dfrac{\pi}{M_c}}{1+\overline{\gamma}}}{2(1+\alpha)\left(1+\overline{\gamma}\sin^2\dfrac{\pi}{M_c}\right)}, & \alpha = 1+r, \\ \dfrac{K_3\left(\dfrac{1}{\overline{\gamma}}\right) - K_3\left(\dfrac{r+2}{\overline{\gamma}(1+\alpha)}\right)}{r+1-\alpha}, & \text{otherwise}. \end{cases}$$

When $M_c = 2$ and 4, (39) reduces to (28) and (34) for BPSK and QPSK, respectively.

IV. Numerical results

Several numerical results are provided in this section to show our new bounds for the outage probability and the error rate performance of GSC in generalized fading channels. These bounds can be used to predict the region in which the GSC performance will be.

Fig. 1 compares the tightness of the upper and the lower bounds (both simple and improved) with the exact outage probability for GSC$(2, 8)$ and GSC$(5, 8)$ in i.i.d. Rayleigh fading. Note that for $M = 2$, the improved upper and lower bounds coincide with the curve corresponding to the exact outage rate of error probability of the GSC$(2, 8)$ receiver. For $M = 5$, the improved upper bound coincides with the exact performance of GSC$(2, 8)$. It is apparent that the lower bounds are much tighter than the upper bounds, particularly when $M \ll L$.

Applying (7), we plot the outage probability of GSC(M, L) in a mixed fading environment as a function of the average SNR of the first branch normalized by the threshold $\gamma^* = \overline{\gamma}_1/\gamma_{\text{th}}$ in Fig. 2. The total number of multipaths is $L = 30$. We assume an exponential multipath intensity profile (MIP) with a decay factor $\delta = 0.7$. The first 15 multipaths are Ricean faded ($K = 3$), the next five multipaths are Rayleigh faded and the remaining 10 multipaths are Nakagami-m faded ($m = 0.75$).

Fig. 3 and Fig. 4 plot the average BER as a function of average SNR of the first branch for BPSK with GSC$(M, 15)$ in i.n.d. Nakagami-m fading ($m = 0.75$) and i.n.d. Ricean fading ($K = 3$), respectively.

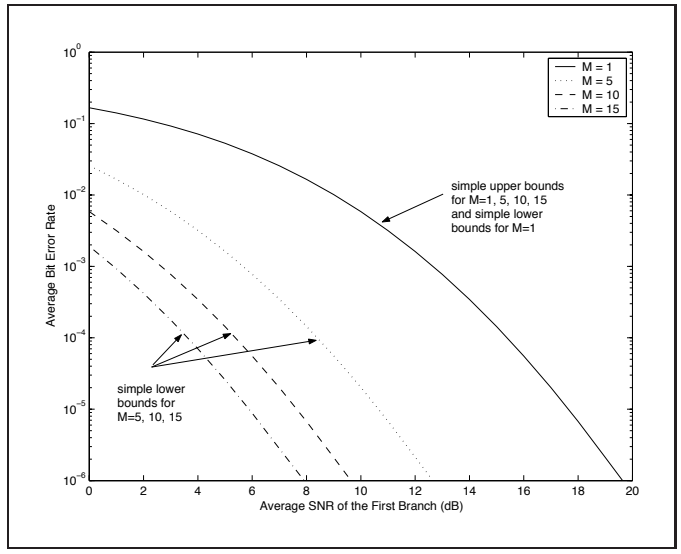

Figure 4: *Simple bounds for the average bit error rate of BPSK with GSC$(M, 15)$ receiver in i.n.d. Ricean fading; $K = 3$.*

The exponential MIP decay factor is $\delta = 0.7$. We readily obtain these curves using Table 2 and (24) with (23).

Fig. 5 compares the tightness of the simple and improved BER bounds for GSC$(2, 10)$ and GSC$(5, 10)$ in i.i.d. Rayleigh fading channels. The lower bounds are much tighter than the upper bounds. As M decreases, the lower bounds become much tighter. In fact, the upper bounds do not depend on M.

V. Conclusion

This paper developed computationally efficient formulas for bounding the GSC performance in independent fading, especially when the order of diversity is large. The mgf of a linear sum of order statistics is needed for performance analysis of GSC. Unfortunately, this mgf does not exist in simple forms for the generalized fading channels. The available mgf expression requires infinite integration, and its computational complexity increases greatly as the diversity order, L, increases. Consequently, a simple expression for the mgf of the GSC output SNR remains an open problem for all but the simplest fading

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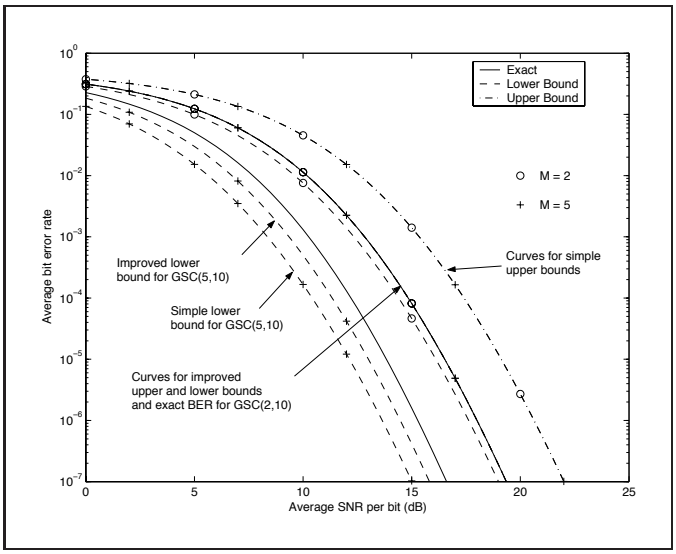

Figure 5: *Error rate of BDPSK with GSC*$(2, 10)$ *and GSC*$(5, 10)$ *in i.i.d. Rayleigh fading.*

distributions (however, see [9]–[10], [16]–[17] for some recent developments concerning this topic). We therefore developed a bounding technique that does not require the mgf. New upper and lower outage bounds were derived for GSC in generalized fading channels. These bounds were further used to derive new bounds for the average error rates of GSC(M, L) in a myriad of fading environments. This facilitates a comparative assessment of various digital modulations with GSC for different fading models and emerging applications.

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